

Transformers and Attention-Based Deep Networks

Term Project

Multi-modal Transformers

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1 Introduction

This is the term project of DI 725: Transformers and Attention-Based Deep Networks. There are four phases in this project. The first phase includes the preparation of a brief literature survey, project proposal and proof of concept. The second phase covers the preliminary results and benchmarking. The third phase is about the results and ablation. The fourth phase concludes the project with discussion, conclusion and presentation.

2 Multi-Modal Transformers

Using the provided remote sensing [dataset](#) (Figure 1), propose an application of transformers and attention-based deep networks. Prepare a summary of the literature and formulate research questions, design benchmarking experiments, obtain results, finalize the ablation study, clean-up the code-base, prepare a report and present the findings.

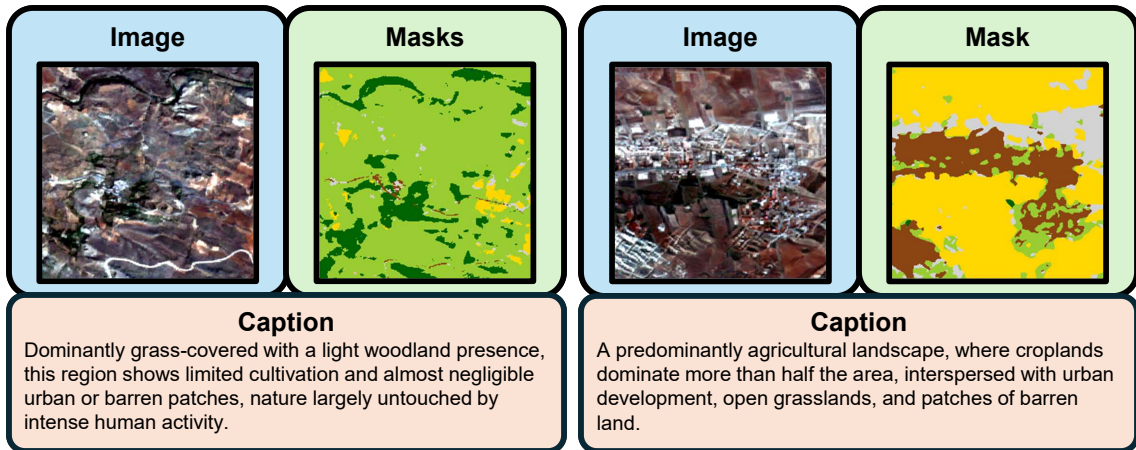


Figure 1: The multi-modal dataset

True color images are available inside *images* folder and respective segmentation maps inside the *masks* folder. There are 7 classes in the segmentation maps, tree cover, scrubland, grassland, cropland, built-up, barren, ice/snow and water and Table 1 shows the RGB values for each class. Descriptions for the images are also available in the *captions* CSV file, with columns *hybrid_gemma3-4b*, *hybrid_qwen3-vl-8b*, *text_qwen3-4b*, *vision_gemma3-4b*, *vision_qwen3-vl-8b*. Columns *Tree*, *Shrub*, *Grass*, *Crop*, *Built-up*, *Barren*, *Water* indicate the composition percentage of each image (derived from segmnetation masks).

Class	R	G	B
Tree	0	100	0
Shrub	255	182	193
Grass	154	205	50
Crop	255	215	0
Built-up	139	69	19
Barren	211	211	211
Water	0	0	255

Table 1: Class RGB Values

3 Deliverables

Submissions are to be uploaded to ODTUCLASS in a single zip file. Do not upload the dataset or any large file (checkpoints, weights, state dictionaries etc.). Uploaded codes must be in .ipynb format, with outputs visible. Check if the

notebook outputs are present. If .py helper scripts were used, the main file must be a notebook (.ipynb) with outputs on. The uploaded report must adhere to the limitations and format described in this document. Failure to comply with deliverable format might result in severe grade deductions.

4 Project Phases

There will be four phases for the term project. It is recommended to start early. All of the project phases will be graded according to the guidelines in their respective section. Deliverables and due dates are available in the following table:

Phase	Tasks	Deliverables	Points	Due Date
1	Literature, Proposal, Proof of Concept	Report and Code	20	5th of April
2	Results, Benchmarking	Report and Code	20	3rd of May
3	Ablation, Discussion, Conclusion	Report and Code	40	7th of June
4	Presentation	Presentation video	20	14th of June

4.1 Phase 1 (20 pts, Due 5th of April)

Prepare a report (1-page maximum) covering literature survey, project proposal and proof of concept. Conduct preliminary experiments and trials with the proposal, to evaluate the feasibility according to the available resources and timeline (proof of concept). Upload the report and proof of concept notebook(s) to ODTUClass.

4.1.1 Literature Survey - 10 pts

Prepare a concise and comprehensive survey of available methods, applications and existing research on multi-modal embeddings. The survey shall be well-supported with appropriate sources. It shall highlight potential gaps and opportunities in the field, leading to the project proposal.

4.1.2 Project Proposal and Proof of Concept - 10 pts

Formulate a research question and prepare a detailed proposal. Identify methods, models, approaches, strategies, performance metrics and other important factors for this project. Even though literature survey points to a solid area within the existing literature, the proposed approach might be infeasible due to constraints and limitations. Design and demonstrate a proof-of-concept, indicating the feasibility of the project proposal.

4.2 Phase 2 (20 pts, Due 3rd of May)

Design benchmarking experiments and obtain experimental results. Extend the report from phase 1 to include the preliminary results (2-page maximum). Explain methods, experiment structure and results in detail. Upload the report and notebook(s) to ODTUClass.

4.2.1 Results - 10 pts

Apply the proposed method and obtain results. Refine the code framework, project plans and address inconsistencies in the training process. The results should solidify the literature and experimentation plans.

4.2.2 Benchmarking - 10 pts

Design ablation experiments to compare the results against a baseline. The goal of benchmarking is to surpass the baseline performance, achieving meaningful improvements. Note: This is not a simple hyperparameter-tuning.

4.3 Phase 3 (40 pts, Due 7th of June)

Finish experiments, conduct ablation study, attain final results. Finalize and submit the report (4-page maximum) including: abstract, introduction, literature, methods, results, discussion and conclusion. Finalize the code-base. Upload the final report and notebook(s) to ODTUClass.

4.3.1 Ablation Study - 10 pts

Finalize and present the ablation results, demonstrating that they are aligned with the research question. A carefully designed ablation study shall cover at least one important change, that will severely affect the results.

4.3.2 Discussion and Conclusion - 10 pts

Discuss the findings, experiments, results, ablation study and compare the results with the literature. Prepare a concise and comprehensive conclusion, clearly highlighting what was accomplished.

4.3.3 Code-base and Report Clarity - 20 pts

Clean up the code-base, add necessary comments, remove unnecessary parts, save the code-base with outputs on as .ipynb notebook. The quality and clarity of the code will be graded. Do not leave magic numbers, unused variables or imports in the final code. The report should be coherent, concise and detailed where necessary.

4.4 Phase 4 (20 pts, Due 14th of June)

Discuss, present and explain the implementation, the code-base and the results. Upload the presentation video to ODTUClass.

4.4.1 Presentation - 20 pts

Record a 10-minute video presentation, explaining the implementation in a clear and structured way, discussing the results and demonstrating the application. Explain the code base, how to run it and then demonstrate how it runs, clearly showing the inputs and outputs. Screen recording or online meeting tools can be used to record this presentation.

5 Appendices

5.1 Research Question

You should write research questions considering the following factors: A good research question should be:

- **Clear and focused.** The question should clearly state what the writer needs to do.
- **Not too broad and not too narrow.** The question should have an appropriate scope. If the question is too broad, it will not be possible to answer it thoroughly within the word limit. If it is too narrow you will not have enough to write about and you will struggle to develop a strong argument.
- **Not too easy to answer.** The question should require more than a simple yes or no answer.
- **Not too difficult to answer.** You must be able to answer the question thoroughly within the given time frame and word limit.
- **Researchable.** You must have access to a suitable amount of quality research materials, such as academic books and refereed journal articles.
- **Analytical rather than descriptive.** Your research question should allow you to produce an analysis of an issue or problem rather than a simple description of it.

We urge you to check these links as they will guide your question-asking and literature search:

- [Developing a research question \[7:23\] University of Melbourne, Academic Skills Youtube Series](#)
- [How to Create Research Questions & A Literature Review \[54:57\] METU Academic Writing Center by Zeynep Ünlüer](#)
- [How to Write a Research Question \(George Mason University\)](#)
- [Identifying a Research Problem and Question, and Searching Relevant Literature \(Daniel J. Boudah, 2011\)](#)

5.2 References

You should not choose a random paper returned from your query in Google Scholar. The paper should have been published in a respectable journal or conference proceeding. You can refer to the [Google Scholar's Journal & Conferences Rank](#) list to find high-impact publications in Artificial Intelligence category. There are many ways to check the quality of the paper. Some of them are:

1. It might be indexed by [Web of Science](#) or [Scopus](#).
2. It might have been published by respectable publishers such as IEEE, ACM, Elsevier, or AAAI.
3. Check the paper or the conference/journal name from [ScienceDirect](#) or [Scimago Journal & Country Rank](#)

You are also expected to reference them properly in your report in IEEE format. All the fields should be complete and proper. Do not rely on what returns from Google Scholar as they are often missing. Check the origin of the paper. Although some appear to be published in Arxiv, they might have been published in a respectable journal or conference afterward. Hence you need to update the details accordingly. If you cite your paper incorrectly or don't use a paper from a good resource (such as one from a predatory journal), you will lose grades accordingly.

5.3 Report Format

Prepare reports in IEEE format. Find the template [here](#) (Conferences, Original Research, Word or Latex).

Final report must include a minimum of one graph and table. The maximum number for both is four. Each figure and table must have a clear and descriptive caption. Make sure the graphs you provide are exported at 300 dpi and not copied and pasted. Failing to do so, or not staying within limits will result in grade reductions.

Results must be shown in tables or other appropriate structures. You can follow the [data-to-viz](#) visualization tool to efficiently and beautifully represent your data and findings. Do not copy and paste code or outputs from your notebook for the paper. This will result in grade deductions. Make sure your graphs, tables, and other material are legible, i.e. readable. Make sure different parts of your graphs (like lines, boxes, and points) can be identified by their shape as well as color. (Like changing line types, markers, etc.) Treat this as if this paper would be printed out in black and white.

Your report will also be evaluated based on adherence to the required format, correctness and sufficient explanation of each section, creation of clear and accurate captions for figures and tables, and references being written completely, accurately, and in the correct format.

5.4 Academic Conduct and Generative AI

Students are expected to uphold the highest standards of academic integrity in all aspects of this project. The use of generative AI tools (e.g., ChatGPT, DALL-E, Copilot, etc.) is permitted under the following conditions:

- Ensure that code generation does not introduce security vulnerabilities and licensing conflicts.
- AI-generated code must be thoroughly reviewed, tested and understood by the author.
- Any code snippet, that is used as-is, should be referenced by either the original contributor or the AI-generator.
- AI-generated visualization shall be reviewed and checked with the original data.
- AI-generated or AI-refined tables and values must be verified with the original data.

You can use AI generation tools for:

Debugging and optimization: Errors are common when working with state-of-the-art and cutting-edge development. Fixes and solutions for different environments can be present in the open-source domain and AI tools are suitable for finding a solution (akin to searching through stackoverflow).

Spell and grammar checking: AI tools can be used to check for spelling and grammar errors.

Code suggestions and completions: AI tools such as ChatGPT and Copilot are suitable for code completion as long as the author knows what and how to implement them.

Visualization creation: Creating visuals from existing data with AI tools can be productive and labor-saving.

Generative data augmentation: AI tools can be employed for generating synthetic data.

Textual drafts: You can generate drafts with AI to adapt to a template.

You should not use AI generation tools for:

Creative content generation: AI tools are infamous for plagiarism and copyright violations.

Finishing entire projects solely with AI tools: Using AI to complete entire implementation tasks without understanding undermines the purpose of the projects and assignments. The projects and assignments are designed to be instructive, real world tasks are often much harder, complex and time-consuming.

Review literature completely with AI: AI tools can be used for summarizing or finding relevant content, however, using only AI for literature review can introduce inherent biases to your work. More importantly, AI tools can hallucinate with the literature content, often leading to irrelevant or sometimes non-existent content.

Substitute yourself with AI tools: Human creativity, consistency and character are irreplaceable with any AI tool, you shall not diminish your effort and contribution by replacing yourself with any AI tool. However, it is beneficial to use AI tools as productivity tools to augment your work.

5.5 General Guidelines for Grading

- The projects should be prepared on an individual basis. You are not allowed to work together. You are expected to follow academic integrity.
- Your whole analyses should be connected/related to each other. You should not do something for the sake of doing it. For instance, if you do apply a feature engineering technique and do not use the results in somewhere else or do not relate it to your research questions, your mark will be degraded accordingly
- Storytelling: You are expected to conduct your analysis by referring to your research questions. You can pose sub-research questions. Check for any spelling or grammar mistakes before the submission of your report. You can use LLMs, Grammarly or Outwrite for this.
- You can refer to ChatGPT ONLY for the purpose of revising and re-writing your paragraphs to enhance the flow. We have tools available to check if you let ChatGPT write nonsense about the topic.
- Your Jupyter notebooks should be self-explanatory and easy to run. We should see all the steps you followed for your analyses in the notebook. You cannot use any other platform (such as Excel or R, even for partial runs) because it will be difficult for us to check their validity.
- The deadline is strict. Hence, start working on your project and documentation as soon as possible. You can't finish it if you leave it to the last minute.
- Before submitting, make sure your notebook runs without errors from start to finish.

5.6 Reproducible Code

Ensuring reproducibility in machine learning projects is crucial for validating results, enabling future improvements, and facilitating collaboration. Follow the guidelines below to make your project easily reproducible.

- The dataset should not be embedded within the submitted Jupyter notebook. Instead, use relative paths when accessing data files (e.g., data/dataset.csv).
- Record all dependencies in requirements.txt to ensure that others can install the necessary packages and replicate your environment.
- To learn more about code reproducibility, you can watch this [presentation](#). Additionally, you can read the following documents: [Organizing Your Projects](#) and [Reproducible Research: Goals, Guidelines and Git](#).
- If you are not familiar with Git and GitHub, you can take the following courses on Datacamp: [Introduction to Git](#) and [Introduction to GitHub Concepts](#). Please reach out to me for access to these courses.
- Even though all the details of your project will be available in your GitHub repository, you must submit the final report, the final Jupyter notebook, and the link to your GitHub repository via ODTUClass.

5.7 Late Submission

Late submission is permitted with 10% penalty for each day, up to 2 days.